

COLUMBIA UNIVERSITY
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STAT 5701

Special Distributions

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Moment Generating Functions

Definition of MGF

Let X be a random variable. Then the moment generating function (mgf) of X, $m_X(t)$, if it exists, is defined as:

$$m_X(t) = E(e^{tX})$$

We say that MGF of X exists, if there exists a positive constant a such that $m_X(t)$ is finite for all $t \in [-a, a]$.

Existence of MGF

Q: Does mgf always exists?

A: No

Main reason: the distribution might have moments $E(X^k)$ which do no exist for some k.

Example: The t distribution does not have a mgf!

Example: The lognormal distribution with pdf

$$f(x) = \frac{1}{\sqrt{2\pi}x} e^{-\frac{(\log x)^2}{2}}, x \geq 0$$

Has finite moments of any order, but its mgf exists only for $t \leq 0$

Note: There are distributions for which only m(0) exists and yet they have all moments finite.

Example 1

Suppose X is a discrete r.v. with

$$p(x) = \begin{cases} \frac{1}{3}, x = 1 \\ \frac{2}{3}, x = 2 \end{cases}$$

Find mgf $m_X(t)$

Property of mgf for discrete rv's

If X is a discrete random variable with a finite range $x_1, \ldots, x_k$, then its mgf is:

$$m(t) = \sum_{i=1}^{k} p(x_i) e^{t x_i}$$

Example: Suppose that the mgf of the random variable X is

$$m(t) = \left(\frac{1 + 2e^t}{3}\right)\left(\frac{1 + 3e^t}{4}\right)$$

Find P(X = 1).

Solution: $m(t) = \frac{(1+2e^t)(1+3e^t)}{12} = \frac{1+3e^t+2e^t+6e^{2t}}{12}$

$$= \frac{1}{12} + \frac{5}{12} e^t + \frac{6}{12} e^{2t} = \frac{1}{12} e^{0 \cdot t} + \frac{5}{12} e^{1 \cdot t} + \frac{1}{2} e^{2 \cdot t}$$

Comparing this expansion to $\sum P(X = x) e^{tx}$:

$$P(X = 0) = \frac{1}{12}, P(X = 1) = \frac{5}{12}, P(X = 2) = \frac{6}{12} = \frac{1}{2}$$

Example 2: MGF of the uniform distribution

Let $X \sim \mathrm{U}(0, 1)$, that is, $f(x) = 1$, $x \in [0, 1]$.

Derive the mgf:

$$m(t) = \int_0^1 e^{xt} \cdot 1 dx = \frac{1}{t} \int_0^1 e^{xt}\, d(xt)$$

$$= \frac{1}{t} e^{xt} \Big]_{x=0}^{1} = \frac{e^t - 1}{t}, t \neq 0$$

Notice that $m(0) = 1$ must be calculated separately!

Final answer:

$$m(t) = \begin{cases} 1, t = 0 \\ \dfrac{e^t - 1}{t}, t \neq 0 \end{cases}$$

Properties of
Moment Generating Functions

1. $m_X(0) = 1$

$$m_X(t) = E\left(e^{tX}\right), \text{ hence } m_X(0) = E\left(e^{0\cdot X}\right) = E(1) = 1$$

2. $$m_X(t) = 1 + \mu_1 t + \frac{\mu_2}{2!}t^2 + \frac{\mu_3}{3!}t^3 + \mathrm{K} + \frac{\mu_k}{k!}t^k + \mathrm{K}$$

3. For all $n \geq 0$, we have that

$$\mu_n = E(X^n) = m_X^{(n)}(0)$$

Where $m_X^{(n)}(0) = \left.\frac{\partial^n m_X(t)}{\partial t^n}\right|_{t=0}$

4. Suppose $X_1, \ldots, X_n$ are n independent random variables, and we defined

$$Y = X_1 + \cdots + X_n$$

Then

$$m_Y(t) = m_{X_1}(t) \cdot \ldots \cdot m_{X_n}(t)$$

MGF not always helpful ☹

Let $X \sim \mathrm{U}(0, 1)$, that is, $f(x) = 1$, $x \in [0, 1]$.

We know $E(X) = \frac{0+1}{2} = \frac{1}{2}$, but we will derive this with the mgf:

$$m(t) = \frac{e^t - 1}{t}, t \neq 0$$

Notice that $m(0) = 1$.

The expected value will be a limit of the derivative:

$$E(X) = m'(0) = \left(\frac{\mathrm{e}^t - 1}{t}\right)' = \frac{te^t - e^t + 1}{t^2} \to \frac{1}{2}$$

as $t \to 0$.

Also note, that you need to use L'Hôpital's rule!

Cumulant Generating Function (CGF)

The cgf provides a way to calculate cumulants, which describe the shape of a distribution and are particularly useful for sums of random variables.

Definition:

$$K_X(t) = \ln\big(M_X(t)\big)$$

Properties:

$$K_X'(0) = E(X)$$
$$K_X''(0) = \mu_2^0 = Var(X)$$
$$K_X'''(0) = \mu_3^0 = E[(X-\mu)^3]$$

Note:

The higher cumulants do not match with the central moments ☹

Bernoulli Distribution

Suppose that we have an experiment that has two outcomes:

- Success (S)
- Failure (F)

These terms are used in reliability testing.

Suppose that p is the probability of success (S) and

$q = 1 - p$ is the probability of failure (F)

This experiment is sometimes called a **Bernoulli Trial**

Define:

$$X = \begin{cases} 0, if\ the\ outcome\ is\ F \\ 1, if\ the\ outcome\ is\ S \end{cases}$$

Then

$$p(x) = P[X = x] = \begin{cases} q & x = 0 \\ p & x = 1 \end{cases}$$

is called the pmf of **Bernoulli** distribution $X \sim \text{Bernoulli}(p)$

Suppose that p is the probability of success (S) and

$q = 1 - p$ is the probability of failure (F)

Now assume that the Bernoulli trial is repeated independently n times, each resulting in either

- Success (S)
- Failure (F)

Let

X = number of successes in the n trials

Q: What is the pmf of X?

Note: the possible values of X are $\{0, 1, 2, \ldots, n\}$

For $n = 5$ the outcomes together with the values of X and the probabilities of each outcome are given in the table below:

FFFFF 0 q^5	*SFFFF* 1 pq^4	*FSFFF* 1 pq^4	*FFSFF* 1 pq^4	*FFFSF* 1 pq^4	*FFFFS* 1 pq^4	*SSFFF* 2 p^2q^3	*SFSFF* 2 p^2q^3
SFFSF *2* p^2q^3	*SFFFS* *2* p^2q^3	*FSSFF* *2* p^2q^3	*FSFSF* *2* p^2q^3	*FSFFS* *2* p^2q^3	*FFSSF* *2* p^2q^3	*FFSFS* *2* p^2q^3	*FFFSS* 2 p^2q^3
SSSFF 3 p^3q^2	*SSFSF* 3 p^3q^2	*SSFFS* 3 p^3q^2	*SFSSF* 3 p^3q^2	*SFSFS* 3 p^3q^2	*SFFSS* 3 p^3q^2	*FSSSF* 3 p^3q^2	*FSSFS* 3 p^3q^2
FSFSS 3 p^3q^2	*FFSSS* 3 p^3q^2	*SSSSF* 4 p^4q	*SSSFS* 4 p^4q	*SSFSS* 4 p^4q	*SFSSS* 4 p^4q	*FSSSS* 4 p^4q	*SSSSS* 5 p^5

x	0	1	2	3	4	5
$p(x) = P[X = x]$	q^5	$5pq^4$	$10p^2q^3$	$10p^3q^2$	$5p^4q$	p^5

For a general n, the outcome of the sequence of n Bernoulli trails is a sequence of **S**'s and **F**'s of length n:

SSFSFFSFFF…FSSSFFSFSFFS

- The value of X for such a sequence is k = the number of **S**'s in the sequence.
- The probability of such a sequence is $p^k q^{n-k}$ (a p for each **S** and a q for each **F**)
- There are $\binom{n}{k}$ such sequences containing exactly k **S**'s
- $\binom{n}{k}$ is the number of ways of selecting the k positions for the **S**'s. (the remaining $n - k$ positions are for the **F**'s

Thus,

$$p(k) = P(X = k) = \binom{n}{k} p^k q^{n-k}, k = 0, \ldots, n$$

Note: These are the terms in the expansion of $(p + q)^n$ using the Binomial Theorem:

$$(p+q)^n = \binom{n}{0} p^0 q^n + \binom{n}{1} p^1 q^{n-1} + \binom{n}{2} p^2 q^{n-2} + \text{K} + \binom{n}{n} p^n q^0$$

For this reason, the probability function is called the probability function for the **Binomial distribution**

Summary

We observe a Bernoulli trial (**S, F**) n times.

Let X denote the number of successes in the n trials. Then X has a **binomial distribution**, i. e.

$$p(x) = P[X = x] = \binom{n}{x} p^x q^{n-x} \quad x = 0,1,2,\text{K}\ ,n$$

Notation:

$$X \sim Binom(n, p)$$

Example 3

A coin is tossed $n = 7$ times. Let X denote the number of **heads (H)** in the $n = 7$ trials.

Then X has a **binomial distribution**, with $p = ½$ and $n = 7$. Thus $X \sim \text{Binom}(7, ½)$ and

$$p(x) = P(X = x) = \binom{n}{x} p^x q^{n-x}, x = 0, \ldots, n$$

$$= \binom{7}{x} \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{7-x}, x = 0, \ldots, 7$$

$$= \binom{7}{x} \left(\tfrac{1}{2}\right)^7 \quad x = 0, 1, 2, \text{K}, 7$$

x	0	1	2	3	4	5	6	7
$p(x)$	$^1/_{128}$	$^7/_{128}$	$^{21}/_{128}$	$^{35}/_{128}$	$^{35}/_{128}$	$^{21}/_{128}$	$^7/_{128}$	$^1/_{128}$

Example 4

If a surgeon performs some surgery, the chance of success is 85%. Suppose that the surgery is performed $n = 20$ times.

Let X denote the number of **successful surgeries** in the $n = 20$ trials.

Then X has a **binomial distribution**, with $p = 0.85$ and $n = 20$.

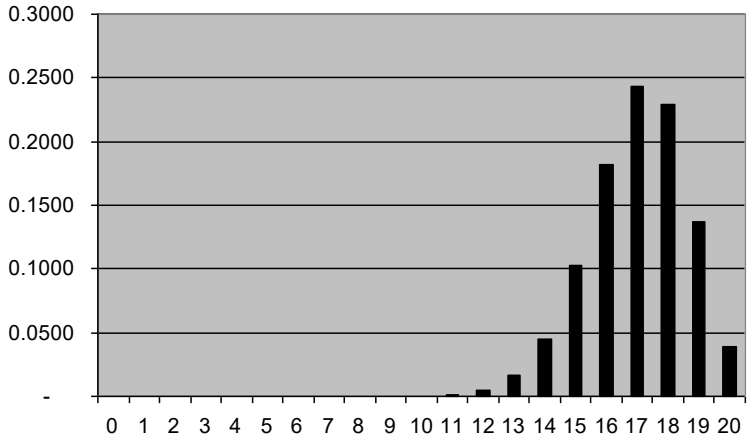
Thus $X \sim$ Binom(20, 0.85) and

$$p(x) = P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

$$= \binom{20}{x} (0.85)^x (0.15)^{20-x}, x = 0, \ldots, 20$$

In Python, you can implement the binomial distribution using the `scipy.stats.binom` module:

`binom.pmf(x, n, p)` $= \binom{n}{x} p^x (1-p)^{n-x}$



x	0	1	2	3	4	5
$p(x)$	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
x	6	7	8	9	10	11
$p(x)$	0.0000	0.0000	0.0000	0.0000	0.0002	0.0011
x	12	13	14	15	16	17
$p(x)$	0.0046	0.0160	0.0454	0.1028	0.1821	0.2428
x	18	19	20			
$p(x)$	0.2293	0.1368	0.0388			

Suppose we want to find the probability that *at least* sixteen operations are successful:

$$= P[X \geq 16]$$

$$= p(16) + p(17) + p(18) + p(19) + p(20)$$

$$= 0.1821 + 0.2428 + 0.2293 + 0.1368 + 0.0388$$

$$= 0.8298$$

Or use software/calculator.

Exercise

A student takes a ten-question true/false exam. Assume the student answers are independent.

a) Find the probability that the student gets exactly six of the questions right simply by guessing the answer on every question.

a) Find the probability that the student will obtain a passing grade of 60% or greater simply by guessing?

Exercise

Suppose that in CA 30% of all homeowners are insured against earthquake damage. Four homeowners are selected at random; let X denote the number among the four who have earthquake insurance.

a) Find the probability distribution of X;

b) Find $P(X = 2) = ?$

c) Draw the corresponding bar plot;

d) What is the most likely value of X (called the mode of the distribution)?

e) What is the probability that at least two of the four selected have earthquake insurance?

Mean and Variance

For the Binomial distribution $X \sim \text{Bin}(n, p)$

$$p(x) = \binom{n}{x} p^x (1-p)^{n-x} \quad x = 0,1,2,\ldots,n$$

Therefore, the moment generating function of X, $m_X(t)$ is:

$$m_X(t) = E\left[e^{tX}\right] = \sum_x e^{tx} p(x)$$

$$= \sum_{x=0}^{n} e^{tx} \binom{n}{x} p^x (1-p)^{n-x}$$

$a^x b^x = (ab)^x$

$$= \sum_{x=0}^{n} \binom{n}{x} \left(e^t p\right)^x (1-p)^{n-x} = \sum_{x=0}^{n} \binom{n}{x} a^x b^{n-x}$$

$$= (a+b)^n = \left(e^t p + 1 - p\right)^n$$

Example: Binomial distribution

We have that $m(t) = (1 - p + pe^t)^n, \forall t \in \mathbb{R}$

Denote q = 1- p. Then:

$$m'(t) = n(q + pe^t)^{n-1}(pe^t)$$

$$\Rightarrow m'(0) = n(q + pe^0)^{n-1}(pe^0) = n(q + p)^{n-1}p$$

$$\Rightarrow \boxed{E(X) = \mu = np}$$

$$m''(t)$$

$$= np[(n-1)(q + pe^t)^{n-2}pe^t e^t + (q + pe^t)^{n-1}e^t]$$

$$= npe^t(q + pe^t)^{n-2}[(n-1)pe^t + q + pe^t]$$

$$= npe^t(q + pe^t)^{n-2}(npe^t + q)$$

$$\Rightarrow m''(0) = np(np + q) = n^2p^2 + npq = E(X^2) = \mu_2$$

$$\Rightarrow \boxed{Var(X) = \mu_2 - \mu_1^2 = npq}$$

Exercise

The Binomial distribution $X \sim \text{Bin}(n, p)$

$$p(x) = \binom{n}{x} p^x (1-p)^{n-x} \qquad x = 0, 1, 2, \text{K}\ , n$$

The moment generating function of X, $m_X(t)$ is:

$$m_X(t) = (1 - p + pe^t)^n$$

Find the cumulant distribution function $K_X(t)$ and the third central moment μ_3^0.

Other discrete distributions

- Geometric distribution
- Poisson distribution
- Negative Binomial distribution
- Hypergeometric distribution

The Geometric distribution

Suppose a Bernoulli trial (S,F) is repeated until a success occurs.

Let X = the trial on which the first success (S) occurs.

Find the probability distribution of X.

Note: the possible values of X are $\{1, 2, 3, 4, 5, \ldots\}$

The sample space for the experiment (repeating a Bernoulli trial until a success occurs is:

$S = \{\text{S, FS, FFS, FFFS, FFFFS, } \ldots \text{, FFF}\ldots\text{FFFS, } \ldots\}$

$$p(x) = P[X = x] = P[\{\overbrace{\text{FFF}\ldots\text{FFF}}^{(x-1)\text{ F's}}\text{S}\}] = (1-p)^{x-1}p$$

Thus, the probability function of X is:

$$P(X = x) = p(x) = p(1 - p)^{x-1} = pq^{x-1},\ x = 1, 2, \ldots$$

A random variable X that has this distribution is said to have the **Geometric** distribution and is denoted $X \sim \text{Geo}(p)$.

Explanation: $p(1) = p$, $p(2) = pq$, $p(3) = pq^2$, $p(4) = pq^3$, ...

forms a geometric series (when added together).

Aside: Geometric series is defined for any a and $|r| < 1$ as

$$a + ar + ar^2 + ar^3 + \ldots = \sum_{k=0}^{\infty} ar^k = \frac{a}{1 - r}$$

Therefore,

$$\sum_{x=1}^{\infty} p(x) = p(1) + p(2) + p(3) + \text{L} \quad = p + pq + pq^2 + pq^3 + \text{L} \quad = \frac{p}{1 - q} = \frac{p}{p} = 1$$

Theorem: The cdf for a geometric random variable X is:

$$F_X(t) = \text{P}(X \le t) = 1 - (1 - p)^{[t]},$$

Where $[t]$ denotes the greatest integer in t.

Example

Suppose a die is rolled until a **six** occurs

Success = **S** = {**six**} , $p = {}^{1}/_{6}$.

Failure = **F** = {**no six**} $q = 1 - p = {}^{5}/_{6}$.

1. What is the probability that it took at most 5 rolls of a die to roll a six?
2. What is the probability that it took at least 10 rolls of a die to roll a six?
3. What is the probability that the "first six" occurred on an even number toss?
4. What is the probability that the "first six" occurred on a toss divisible by 3 given that the "first six" occurred on an even number toss?

Solution

Let X denote the toss on which the first head occurs.

Then X has a geometric distribution with $p = {}^{1}/_{6}$.. $q = 1 - p = {}^{5}/_{6}$.

$$P[X = x] = p(x) = pq^{x-1} = \left(\tfrac{1}{6}\right)\left(\tfrac{5}{6}\right)^{x-1} \quad x = 1, 2, 3, \text{K}$$

1. $P[X \leq 5]$?
2. $P[X \geq 10]$?
3. $P[X$ is divisible by $2]$?
4. $P[X$ is divisible by $3 | X$ is divisible by $2]$?

1. $P[X \le 5]$?

$$P\left[X \le 5\right] = p(1) + p(2) + p(3) + p(4) + p(5)$$

$$= \left(\tfrac{1}{6}\right)\left(\tfrac{5}{6}\right)^0 + \left(\tfrac{1}{6}\right)\left(\tfrac{5}{6}\right)^1 + \left(\tfrac{1}{6}\right)\left(\tfrac{5}{6}\right)^2 + \left(\tfrac{1}{6}\right)\left(\tfrac{5}{6}\right)^3 + \left(\tfrac{1}{6}\right)\left(\tfrac{5}{6}\right)^4$$

$$= \left(\tfrac{1}{6}\right)\left[1 + \left(\tfrac{5}{6}\right)^1 + \left(\tfrac{5}{6}\right)^2 + \left(\tfrac{5}{6}\right)^3 + \left(\tfrac{5}{6}\right)^4\right] = \left(\tfrac{1}{6}\right)\left[\frac{1 - \left(\tfrac{5}{6}\right)^5}{1 - \tfrac{5}{6}}\right] = 1 - \left(\tfrac{5}{6}\right)^5$$

Using $a + ar + ar^2 + \mathrm{K} + ar^{n-1} = a\dfrac{1 - r^n}{1 - r}$

$$a + ar + ar^2 + \mathrm{K} = a\frac{1}{1 - r}$$

Or directly: $F(5) = 1 - (1\text{-}1/6)^{[5]} = 1 - (5/6)^5$

2. $P(X \ge 10) = 1 - P(X \le 9) = 1 - \left[1 - \left(\tfrac{5}{6}\right)^9\right] = \left(\tfrac{5}{6}\right)^9$

3. $P[X \text{ is divisible by } 2]$?

$$P\left[X \text{ is divisible by } 2\right] = p(2) + p(4) + p(6) + \text{K}$$

$$= \left(\tfrac{1}{6}\right)\left(\tfrac{5}{6}\right)^1 + \left(\tfrac{1}{6}\right)\left(\tfrac{5}{6}\right)^3 + \left(\tfrac{1}{6}\right)\left(\tfrac{5}{6}\right)^5 + \text{K}$$

$$= \left(\tfrac{1}{6}\right)\left(\tfrac{5}{6}\right)\left[1 + \left(\tfrac{5}{6}\right)^2 + \left(\tfrac{5}{6}\right)^4 + \text{K}\right] = \left(\tfrac{1}{6}\right)\left(\tfrac{5}{6}\right)\left[\frac{1}{1 - \left(\tfrac{5}{6}\right)^2}\right]$$

$$= \tfrac{5}{36}\left[\frac{1}{1 - \tfrac{25}{36}}\right] = \tfrac{5}{36-25} = \tfrac{5}{11}$$

a

r

in geom. series

4. $P(X \text{ is divisible by } 3 \mid X \text{ is divisible by } 2) = ?$

Solution:

$$P(X \text{ is divisible by } 3 \mid X \text{ is divisible by } 2)$$
$$= \frac{P(\{X \text{ is divisible by } 3\} \cap \{X \text{ is divisible by } 2\})}{P(X \text{ is divisible by } 2)}$$
$$= \frac{P(X \text{ is divisible by } 6)}{P(X \text{ is divisible by } 2)}$$

Now,

$$P(X \text{ is divisible by } 6) = p(6) + p(12) + p(18) + \ \ldots$$
$$= \left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^5 + \left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^{11} + \left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^{17} + \cdots$$
$$= \left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^5 \left[1 + \left(\frac{5}{6}\right)^6 + \left(\frac{5}{6}\right)^{12} + \cdots\right]$$
$$= \left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^5 \frac{1}{1 - \left(\frac{5}{6}\right)^6} = \frac{5^5}{6(6^6 - 5^6)} = \frac{3125}{31031}$$

Hence

$$P\left[\, X \text{ is divisible by } 3 \,\middle|\, X \text{ is divisible by } 2 \,\right]$$

$$= \frac{P(X \text{ is divisible by } 6)}{P(X \text{ is divisible by } 2)} = \frac{\frac{3125}{31031}}{\frac{5}{11}} = \frac{625}{2821}$$

The Poisson Distribution

- Suppose events are occurring sequentially in time.
- Let X be the number of events occurring in a fixed period. Then X will have a **Poisson distribution** with parameter λ
- $p(x) = \frac{\lambda^x}{x!} e^{-\lambda}, x = 0, 1, 2, 3, \ldots$

□ λ is called the intensity parameter

Some properties of the probability function for the **Poisson distribution** with parameter λ.

$$\sum_{x=0}^{\infty} p(x) = \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} e^{-\lambda} = 1$$

$$\sum_{x=0}^{\infty} \frac{\lambda^x}{x!} e^{-\lambda} = e^{-\lambda}\left(1+\lambda+\frac{\lambda^2}{2!}+\frac{\lambda^3}{3!}+\frac{\lambda^4}{4!}+\mathrm{K}\right)$$

$$= e^{-\lambda}\left(e^{\lambda}\right) = 1$$

using $\quad e^{u} = 1+u+\frac{u^2}{2!}+\frac{u^3}{3!}+\frac{u^4}{4!}+\mathrm{K}$

Example

The number of hurricanes over a period of a year in the Caribbean is known to have a Poisson distribution with $\lambda = 13.1$

Determine the probability function of X.

Compute the probability that X is at most 8.

Compute the probability that X is at least 10.

Given that at least 10 hurricanes occur, what is the probability that X is at most 15?

Solution

- X will have a **Poisson distribution** with parameter $\lambda = 13.1$, i.e.

$$p(x) = \frac{\lambda^x}{x!} e^{-\lambda} \qquad x = 0, 1, 2, 3, 4, \text{K}$$

$$= \frac{13.1^x}{x!} e^{-13.1} \qquad x = 0, 1, 2, 3, 4, \text{K}$$

Table of $p(x)$

x	$p(x)$	x	$p(x)$
0	0.000002	10	0.083887
1	0.000027	11	0.099901
2	0.000175	12	0.109059
3	0.000766	13	0.109898
4	0.002510	14	0.102833
5	0.006575	15	0.089807
6	0.014356	16	0.073530
7	0.026866	17	0.056661
8	0.043994	18	0.041237
9	0.064036	19	0.028432

$$P[\text{at most } 8] = P[X \le 8]$$

$$= p(0) + p(1) + \text{L} + p(8) = .09527$$

$$P[\text{at least } 10] = P[X \ge 10] = 1 - P[X \le 9]$$

$$= 1 - \left(p(0) + p(1) + \text{L} + p(9) \right) = .8400$$

$$P\left[\text{at most } 15 \middle| \text{at least } 10 \right] = P\left[X \le 15 \middle| X \ge 10 \right]$$

$$= \frac{P\left[\{X \le 15\} \cap \{X \ge 10\} \right]}{P[X \ge 10]} = \frac{P[10 \le X \le 15]}{P[X \ge 10]}$$

$$= \frac{p(10) + p(11) + \text{L} + p(15)}{.8400} = 0.708$$

2. Poisson approximation to Binomial

$$p_{Bin}\left(x\middle|p,n\right)=\binom{n}{x}p^{x}\left(1-p\right)^{n-x}$$

is the probability function for the Binomial distribution with parameters n and p.

If we let $n \to \infty$ and $p \to 0$ such that $np = \lambda$, then

$$\lim_{n\to\infty,\,p\to 0} p_{Bin}\left(x\middle|p,n\right)=p_{Poisson}\left(x\middle|\lambda\right)=\frac{\lambda^{x}}{x!}e^{-\lambda}$$

Example

Five hundred people are attending a meeting. Approximate the probability that at most one of the five hundred was born on September 1st .

Example Solution

Five hundred people are attending a meeting. Approximate the probability that at most one of the five hundred was born September 1st .

We have: $n = 500, p = \frac{1}{365}$

Exact answer with Binomial:

$$P(X \leq 1) = \binom{n}{0} p^0 q^n + \binom{n}{1} p^1 q^{n-1} = 0.6021046$$

Approx.: *X* is Poison with:

$$\lambda = np = \frac{500}{365} = 1.37$$

$$\Rightarrow P(X \leq 1) = \frac{\lambda^0 e^{-\lambda}}{0!} + \frac{\lambda^1 e^{-\lambda}}{1!} = e^{-1.37} + 1.37(e^{-1.37}) = 0.6022812$$

Exercise

Let X be a Poisson random variable with parameter λ. Show that the probability that X is even is:

$$\frac{1}{2}\left(1 + e^{-2\lambda}\right)$$

Hint:

$$e^{\lambda} + e^{-\lambda} = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} + \sum_{k=0}^{\infty} \frac{(-\lambda)^k}{k!} = 2\sum_{k=0}^{\infty} \frac{\lambda^{2k}}{(2k)!}$$

Negative Binomial Distribution

Suppose a Bernoulli trial (S,F) is repeated until k successes occur.

Let X = the trial on which the kth success (S) occurs.

Find the probability distribution of X.

Note: the possible values of X are

$$\{k,\ k+1,\ k+2,\ k+3,\ k+4,\ k+5,\ \ldots\ \}$$

The sample space for the experiment (repeating a Bernoulli trial until k successes occurs) consists of sequences of S's and F's having the following properties:

1. each sequence will contain k S's

2. The last outcome in the sequence will be an S.

A sequence of length x containing exactly k S's

SFSFSFFFFS FFFSF … FFFFFFS

The # of S's in the first $x-1$ trials is $k-1$.

The last outcome is an S

$$p(x) = P[X = x] = \binom{x-1}{k-1} \underbrace{p^k q^{x-k}} \qquad x = k, k+1, k+2, \text{K}$$

The # of ways of choosing from the first $x-1$ trials, the positions for the first $k-1$ S's.

The probability of a sequence containing k S's and $x-k$ F's.

Example

Suppose the chance of winning any prize in a lottery is 3%. Suppose that I play the lottery until I have won $k = 5$ times.

Let X denote the number of times that I play the lottery.

Find the probability function, $p(x)$, of X

$$p(x) = P[X = x] = \binom{x-1}{k-1} p^k q^{x-k} \qquad x = k, k+1, k+2, \text{K}$$

$$= \binom{x-1}{4} (0.03)^5 (0.91)^{x-5} \qquad x = 5, 6, 7, \text{K}$$

Hypergeometric distribution

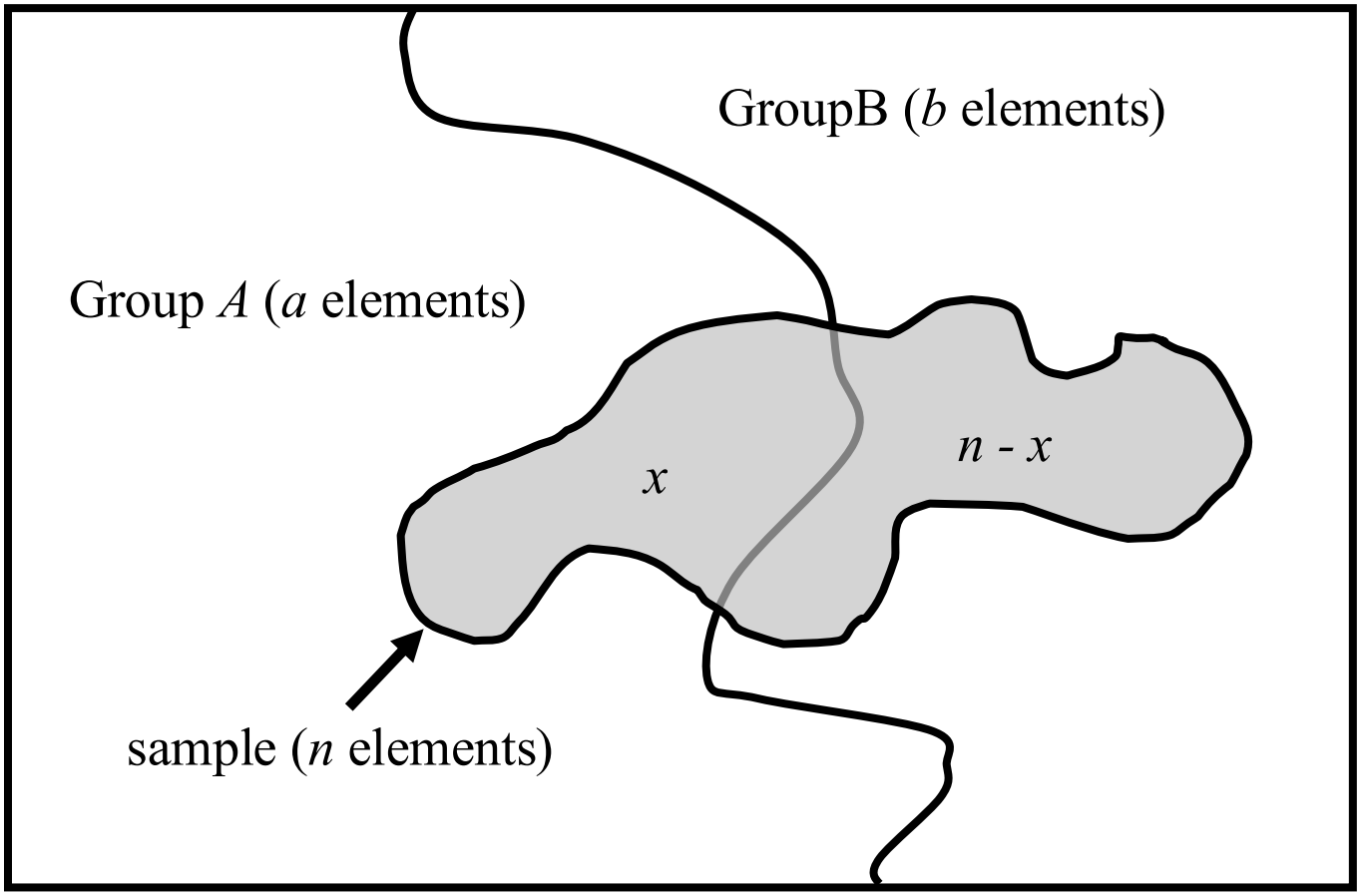
Suppose we have a population containing N objects.

Suppose the elements of the population are partitioned into two groups. Let a = the number of elements in group A and let b = the number of elements in the other group (group B). Note $N = a + b$.

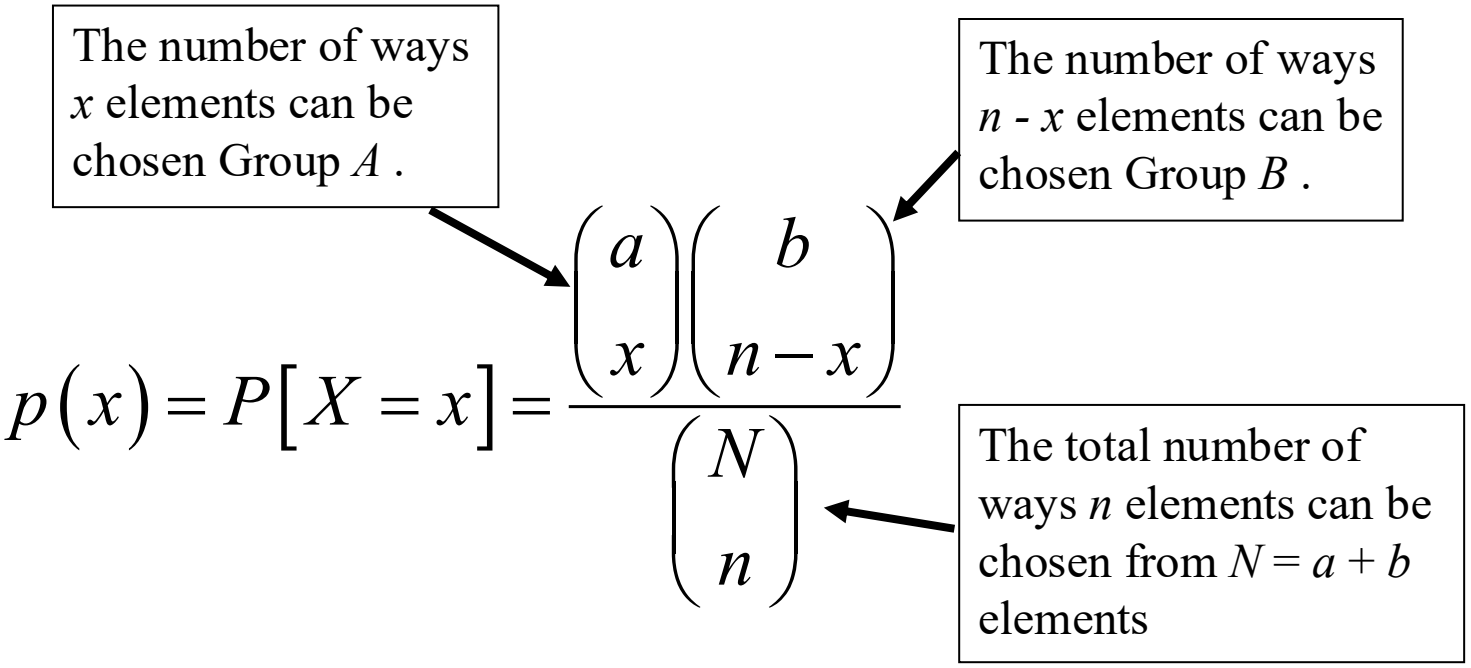
Now suppose that n elements are selected from the population at random. Let X denote the elements from group A. ($n - X$ will be the number of elements from group B.)

Find the probability distribution of X.\

Population



Thus, the pmf of X is:



A random variable X that has this distribution is said to have the **Hypergeometric distribution**.

The possible values of X are integer values that range is:

$$\max\{0, n-b\} \le x \le \min\{n, a\}$$

Example: Hypergeometric distribution

Suppose that $N = 10$ automobiles have just come off the production line. Also assume that $a = 3$ are defective (have serious defects). Thus $b = 7$ are defect-free.

A sample of $n = 4$ are selected and tested to see if they are defective. Let X = the number in the sample that are defective. Find the probability function of X.

From the above discussion X will have a hypergeometric distribution with

$$p(x) = \frac{\binom{3}{x}\binom{7}{4-x}}{\binom{10}{4}}, x = 0, 1, 2, 3$$

x	$p(x)$
0	0.1667
1	0.5000
2	0.3000
3	0.0333

In R: use `dhyper(x, 3, 7, 4)`

Sampling with and without replacement

Suppose we have a population containing N objects.

Suppose the elements of the population are partitioned into two groups. Let a = the number of elements in group A and let b = the number of elements in the other group (group B). Note $N = a + b$.

Now suppose that n elements are selected from the population at random. Let X denote the elements from group A. ($n - X$ will be the number of elements from group B.)

Find the probability distribution of X.

1. If the sampling was done with replacement.

2. If the sampling was done without replacement

Solution:

1. If the sampling was done with replacement.

Then the distribution of X is the **Binomial distn.**

with $p = \frac{a}{N}$ and $q = 1 - p = \frac{b}{N}$

i.e. $p_{Binom}(x) = P[X = x] = \binom{n}{x}\left(\frac{a}{N}\right)^{x}\left(\frac{b}{N}\right)^{n-x}$

2. If the sampling was done without replacement.

Then the distribution of X is the **hypergeometric distn.**

i.e. $p_{Hyper}(x) = P[X = x] = \frac{\binom{a}{x}\binom{b}{n-x}}{\binom{N}{n}}$

Result:

$$p_{Hyper}(x) = \binom{a}{x}\binom{b}{n-x} \Big/ \binom{N}{n} \cong$$

$$p_{Binom}(x) = \binom{n}{x}\left(\frac{a}{N}\right)^{x}\left(\frac{b}{N}\right)^{n-x}$$

for large values of N, a and b

Thus for large values of N, a and b sampling **with replacement** is equivalent to sampling **without replacement**.

The Multinomial distribution

Suppose that we observe an experiment that has k possible outcomes $\{O_1, O_2, \ldots, O_k\}$ independently n times.

Let $p_1, p_2, \ldots, p_k$ denote probabilities of $O_1, O_2, \ldots, O_k$ respectively.

Let X_i denote the number of times that outcome O_i occurs in the n repetitions of the experiment.

Then the joint probability function of the random variables $X_1, X_2, \ldots, X_k$ is

$$p(x_1, \mathrm{K}\ , x_n) = \frac{n!}{x_1!x_2!\mathrm{K}\ x_k!} p_1^{x_1} p_2^{x_2} \mathrm{K}\ \ p_k^{x_k}$$

Note:

$$p_1^{x_1} p_2^{x_2} \mathrm{K}\ p_k^{x_k}$$

is the probability of a sequence of length n containing

x_1 outcomes O_1

x_2 outcomes O_2, … , x_k outcomes O_k

$$\frac{n!}{x_1!x_2!\mathrm{K}\ x_k!} = \binom{n}{x_1\quad x_2\quad \mathrm{L}\quad x_k}$$

is the number of ways of choosing the positions for the x_1 outcomes O_1, x_2 outcomes O_2, …, x_k outcomes O_k

$$\binom{n}{x_1}\binom{n-x_1}{x_2}\binom{n-x_1-x_2}{x_3}\mathrm{K}\binom{x_k}{x_k}$$

$$= \left(\frac{n!}{x_1!(n-x_1)!}\right)\left(\frac{(n-x_1)!}{x_2!(n-x_1-x_2)!}\right)\left(\frac{(n-x_1-x_2)!}{x_3!(n-x_1-x_2-x_3)!}\right)\mathrm{K} = \frac{n!}{x_1!x_2!\mathrm{K}\ x_k!}$$

Summary:

$$p(x_1, \ldots, x_n) = \frac{n!}{x_1!\, x_2! \ldots x_k!} p_1^{x_1} p_2^{x_2} \ldots p_k^{x_k}$$

$$= \binom{n}{x_1 \quad x_2 \quad \mathrm{K} \quad x_k} p_1^{x_1} p_2^{x_2} \mathrm{K}\ p_k^{x_k}$$

is called the **Multinomial** distribution

Example:

Suppose that a treatment for back pain has three possible outcomes:

O_1 - Complete cure (no pain) – (30% chance)

O_2 - Reduced pain – (50% chance)

O_3 - No change – (20% chance)

Hence $p_1 = 0.30$, $p_2 = 0.50$, $p_3 = 0.20$.

Suppose the treatment is applied to $n = 4$ patients suffering back pain and let X = the number that result in a complete cure, Y = the number that result in just reduced pain, and Z = the number that result in no change.

Find the distribution of X, Y and Z. Compute $P[X + Y \geq Z]$

$$p(x,y,z) = \frac{4!}{x!\,y!\,z!}(0.30)^x (0.50)^y (0.20)^z \quad x + y + z = 4$$

$$P[X+Y \geq Z] = 0.9728$$

x	y	z				
		0	1	2	3	4
0	0	0	0	0	0	0.0016
0	1	0	0	0	0.0160	0
0	2	0	0	0.0600	0	0
0	3	0	0.1000	0	0	0
0	4	0.0625	0	0	0	0
1	0	0	0	0	0.0096	0
1	1	0	0	0.0720	0	0
1	2	0	0.1800	0	0	0
1	3	0.1500	0	0	0	0
1	4	0	0	0	0	0
2	0	0	0	0.0216	0	0
2	1	0	0.1080	0	0	0
2	2	0.1350	0	0	0	0
2	3	0	0	0	0	0
2	4	0	0	0	0	0
3	0	0	0.0216	0	0	0
3	1	0.0540	0	0	0	0
3	2	0	0	0	0	0
3	3	0	0	0	0	0
3	4	0	0	0	0	0
4	0	0.0081	0	0	0	0
4	1	0	0	0	0	0
4	2	0	0	0	0	0
4	3	0	0	0	0	0
4	4	0	0	0	0	0

Summary

Discrete distributions

Discrete distributions

Distribution	PMF	Support	Mean E[X]	Variance Var(X)
Bernoulli	$p(x) = p^x(1-p)^{1-x}$	$x \in \{0,1\}$	p	$p(1-p)$
Binomial	$p(x) = \binom{n}{x}p^x(1-p)^{n-x}$	$x \in \{0,1,2,\dots,n\}$	np	$np(1-p)$
Poisson	$p(x) = \frac{\lambda^x e^{-\lambda}}{x!}$	$x \in \{0,1,2,\dots\}$	λ	λ
Geometric	$p(x) = (1-p)^{x-1}p$	$x \in \{1,2,3,\dots\}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Negative Binomial	$p(x) = \binom{x-1}{r-1}p^r(1-p)^{x-r}$	$x \in \{r, r+1, r+2, \dots\}$	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$
Discrete Uniform	$p(x) = \frac{1}{b-a+1}$	$x \in \{a, a+1, \dots, b\}$	$\frac{a+b}{2}$	$\frac{N^2-1}{12}$
Hypergeometric	$p(x) = \frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}$			